Prakhar Singh

Junior Research Fellow (JRF)
Aryabhatta Research Institute of Observational Sciences (ARIES)

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Research Area

Solar Coronal Jets and Associated Small-Scale Flaring Processes During Solar Cycles 24 and 25.
My research investigates the origin and evolution of solar coronal jets and associated small-scale flaring using high-resolution data from EUI/Solar Orbiter, IRIS, AIA/SDO, MAST, e-Callisto, XSM/Chandrayaan-2, and ADITYA-L1. I also perform coronal magnetic field modeling to study the magnetic topology around these jets, aiming to enhance understanding of solar dynamics and space weather.

Academic Qualification

- **Aryabhatta Research Institute of Observational Sciences (ARIES)** — *Dec 2023 - ongoing*
  PhD — Percentage: 85.5 %
- **Department of Physics, Banaras Hindu University** — *2021-2023*
  MSc Physics — CGPA: 8.73 / 10
- **Department of Physics, Aligarh Muslim University** — *2018-2021*
  BSc Physics (Hons.) — CGPA: 8.15 / 10

Research Projects

- **Energetic and Kinematic Properties of Solar Active Region Jets** — *May 2024- ongoing*
  Supervisor - Dr. Vaibhav Pant | ARIES
  - Studied recurrent solar jets using SDO/AIA and IRIS data, analyzing their dynamics and energy contents.
  - Calculated kinematic properties (e.g., length, width, duration, velocity) and estimated emission measure, mean temperature, and electron density using DEM analysis.
  - Estimated energy flux components (kinetic, potential, enthalpy, and radiative flux).
- **Chromospheric Heating during an active region evolution** — *June 2023 - Dec 2023*
  Supervisor - Dr. Sreejith Padinhatteeri | Manipal Center of Natural Sciences, MAHE
  - Explored the correlation between an active region's UV, EUV emission, and magnetic activity.
  - DEM analysis using the Hannah and Kontar method to derive temperature and emission maps from SDO/AIA's six EUV channels.
- **Remote Sensing of Lower Ionosphere using Very Low Frequency Wave** — *Aug 2022 - April 2023*
  Final Year M.Sc. Project | Supervisor - Prof. Abhay Kumar Singh
  - Investigated the night-time D-layer height, electron density and propagation distance of the waves.
  - Analyzed the diurnal and seasonal variations of Tweek occurrence in the South Asian region using data from the Automatic Whistler Detector.

Skills

- **Programming Languages:** - Python, IDL, Fortran
- **Professional Softwares:** - OriginPro, SAO Image DS9, IRAF, VIREO, Aladin
- **Operating System:** Linux, Windows
- **Drawing & Typesetting** LATEX, MS Office
- **Communication:** Conference presentations, poster preparation, academic writing
- **Languages** English, Hindi (Native)

Workshop/Conference/School Attended

- Solar Cycle Variability Conference 2024: From understanding to making prediction jointly organized by IIT BHU and ARIES, Nainital | 14-18 Oct 2024
- 2nd Workshop on Space Weather Science and Opportunities followed by the 3rd Indian Space Weather Conference (ISWC-3) organized by the IIT Roorkee | 5-9 Oct 2024.
- 7th Aditya L1 Support cell Workshop held at ARIES | 14-18 Oct 2024
- 5th Aditya L1 Support cell Workshop held at IIT Kanpur conducted by ARIES and ISRO
- Exploring the Sun and Heliosphere using Aditya L1 national Workshop conducted by ARIES and IIT BHU | 2-5 Feb 2023
- Design of Space Payloads and Developmental Approach conducted by Manipal Centre of Natural Sciences, MAHE | June 2023
- IMPRESS 2023 (Inspiring the Minds of Post-graduates for Research in Earth and Space Science) organized by Indian Institute of Geomagnetism, Dept. of Science and Technology. | 11-14 Feb 2023
- 11th IIST Astronomy and Astrophysics School organized by Indian Institute of Space science and Technology. | 14-23 Dec 2022
- 1st Indo-US Collaborative Meeting Space Radiation Workshop organized by ARIES, | 24-28 Jan 2022
- 2022 Qiskit Global Summer School on Quantum Simulation
- 2021 Qiskit Global Summer School on Quantum Machine Learning

Awards/Achievements

- Secured 3rd rank in Master of Science (Physics) class of 2023 at Banaras Hindu University.
- Ranked in the top 5 % of the Introduction to Research course offered by IIT Madras (NPTEL).
- Ranked 4th in the Electronic Theory of Solids course offered by IIT Kharagpur (NPTEL).
- University topper of National Graduate Physics Examination (NGPE-2021) organized by Indian Association of Physics Teachers (IAPT).
- Student Ambassador of International Astronomy and Astrophysics Competition IAAC-20
- Best Student Ambassador at National Students Space Challenge (NSSC) IIT Kharagpur (2019)

Short Courses

- Data- driven Astronomy an online course by The University of Sydney through Coursera.
- Analyzing the Universe, an online course by Rutgers the State University of New Jersey through Coursera.
- Understanding Research Methods" an online course offered by University of London through Coursera.
- Remote Sensing and GIS applications in Atmospheric and Oceanic Hazards, an online course by IIRS, ISRO, Department of Space.
- Introduction to Research, 3 months online course offered by IIT Madras through NPTEL.
- Introduction to Classical Mechanics, 3 months online course offered by IIT Hyderabad through NPTEL.
- Electronic Theory of Solids , 3 months online course offered by IIT Kharagpur through NPTEL.
- Satellite Photogrammetry and its Application, an online course offered by IIRS, ISRO, Department of Space.
- Advanced Course of Special Theory of Relativity online course by Prof. H.C Verma.
- Learning Physics through Simple Experiments online course by Prof. H.C Verma.

Coursework

- Solar Physics
- Plasma Physics and Magnetohydrodynamics
- Mathematical and Statistical Methods
- Electrodynamics and Radiative Processes
- Astrophysical Fluid Dynamics
- Observational Techniques and Methodology
- Formation, Evolution and Atmosphere of stars
- Interstellar medium and Stellar variability